Table 37. From reference implementation to production

Capability	Reference implementation	Production replacement	When
Corpus storage	JSON file	PostgreSQL catalogue and S3-compatible object store	Months 1 to 6
Ingestion	Markdown loader	PDF and HTML extraction with Bangla OCR and quality scoring	Months 1 to 9
Lexical search	In-memory BM25	OpenSearch or PostgreSQL full-text search	Months 4 to 9
Dense retrieval	Not included	Multilingual embeddings in a vector index; hybrid ranking	Months 6 to 12
Answers	Extractive with citations	Extractive first; optional open-weight LLM rephrasing behind citation verification	After G2 evidence
API	Standard-library HTTP server	API gateway with authentication, quotas and versioning	Months 6 to 9
Identity	None	Single sign-on with MFA and roles	Before G3
Audit log	Hash-chained JSONL file	Append-only store with replication and external anchoring	Before G3
Register and risk tool	CLI and JSON files	Governance workbench web application	Months 9 to 15

Source: BAPIL Technical Blueprint, Version 1.0 (2026). CC BY 4.0.